Bingi Gnaneswanth

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Bingi_Gnaneswanth

SUMMARY

Computer Science student with hands-on project experience in Java backend development, distributed systems, and machine learning. Built REST APIs, a fault-tolerant distributed system, and projects involving chest X-ray image processing and AI model evaluation. Interested in building reliable AI systems and solving real-world problems through technology.

SKILLS

Languages: Java,Python,SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Backend: Spring Boot, Spring MVC, REST APIs, gRPC, Protocol Buffers, Node.js

Frontend: React, TypeScript, HTML, CSS

Databases: MySQL, PostgreSQL, MongoDB, RocksDB

AI/ML: Contextual Bandits, LinUCB, Online Learning, Off-Policy Evaluation, Policy Optimization, Machine Learning, RAG, FAISS, Sentence Transformers, ChromaDB

Tools: Git, Maven, Docker, Prometheus, Grafana

PROJECTS

Distributed Fault-Tolerant Key-Value Store using the Raft Consensus Algorithm

- Built a **fault-tolerant distributed key-value store** implementing Raft **leader election, log replication, quorum-based consensus, and automatic failover** for strong consistency.
- Implemented **RocksDB persistence, gRPC-based inter-node communication, crash recovery, and snapshotting** for reliable distributed state management.
- Developed a **React real-time cluster dashboard** and instrumented **25+ Prometheus metrics** for node health, consensus events, replication latency, RPC performance, and KV operations.
- **Containerized the multi-node cluster with Docker** and integrated **Prometheus/Grafana** for real-time observability, performance monitoring, and failure analysis.

Medical Image Super-Resolution for Chest X-Rays

- Developed **4× chest X-ray super-resolution models** using SwinIR and RRDB architectures.
- Compared **VGG and CheXNet perceptual losses** through a 12-run ablation study across four configurations.
- Evaluated reconstruction quality using **PSNR and LPIPS** and diagnostic performance using **ROC-AUC**, finding no statistically significant AUC improvement.

EDUCATION

Dayananda Sagar University Sep 2023 – Ongoing 2027 | Bangalore
Bachelor of Technology in Computer Science Engineering

Gitam Junior College Jun 2022 – Apr 2023 | Tirupati
MPC-Mathematics, Physics, Chemistry